



Analyzing Breast Cancer Data to Better Understand Patient Diagnosis and Outlook

Team 3

Rassul Safi, Nishan Ghale, Gnaneshwar Reddy Vemunuri, Venkata Naga
Sanjay Reddy Narra, Satya Durgesh Keerthi, Sai Shree Varshitha
Lingamneni, Kishore Karanam

Agenda

Introduction

Proposed Data Analysis

Model Building

Model Evaluation

Results

Recommendations

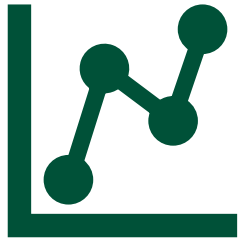
Future Research Directions

Conclusion

References



INTRODUCTION



Big Data Analytics can be leveraged into any domain to help gain insight.

Improve operations, make decisions, understand occurrences.



Selected Domain: Healthcare

Vast amount of data.

Insights can be applied.

Problem Statement

Determine patient outlook through tumor analysis.

Benign or malignant?

Analyze size, texture, perimeter, and other physical characteristics.

Our Machine Learning Model will:

Analyze tumor properties to give faster diagnosis.

Improve patient care with faster, more accurate diagnoses.

Motivations

Improve healthcare services.

Provide healthcare providers with a legitimate tool.

Research Questions and Hypothesis



What tumor characteristics are most significant in predicting whether a tumor is malignant or benign?

Tumor texture and perimeter correlate with malignant tumors, tumor area and concavity will correlate with benign tumors.



Is there a significant correlation between certain tumor attributes (such as smoothness and concavity) and the likelihood of malignancy?

Yes correlations between attributes such as perimeter and smoothness in relation to malignancy.



What are the most and least important tumor attributes in determining whether a tumor is malignant or benign?

Important malignancy prediction: Tumor perimeter and smoothness.

Less Important: Perimeter and smoothness.

Problem Description

Significance of Healthcare Data:

Enhances patient care by using data analysis to produce actionable insights.

Understanding complicated illnesses like breast cancer is aided by machine learning.

Breast Cancer Complexity:

Patients differ greatly in the size, shape, and texture of their tumors.

Forecasting results and directing individualized treatments are made easier with insights.

Impact of Machine Learning:

Identifies if tumor traits linked to malignancy or benignity.

Enhances patient survival chances by providing information for treatment plans.

Improves early detection and precise diagnoses.

Broader Implications:

Improved screening methodologies.

Easier identification of high-risk patients for intensive therapy or monitoring.

Selected Datasets

Data source

- Retrieved from Hugging Face platform
- Originated from Wisconsin Diagnostic Breast Cancer dataset

Key Characteristics

- Focuses on patient data for breast cancer
- Various types of data included: nominal, ordinal, and ratio

Attributes

- Attributes contain ID, Diagnosis, Radius, Texture, Perimeter, Area, Smoothness, Compactness, Concavity, Symmetry, Fractal Dimensions.

Literature Review

Breast Cancer:

- Affects breast cancer tissue.
- Second most common form of cancer for women in the United States (“Breast Cancer,” 2024).

Current Diagnosis Tools:

- Mammograms, breast exams, MRI’s, and ultrasounds.

R Package:

- TumGroth

AI Tools:

- Clearer images

Proposed Data Analysis



Pre-processing: data collecting and data cleaning before exploratory data analysis and classification are performed.



Different data analysis technologies: Python and a variety of libraries such as Pandas, NumPy, matplotlib, SciPy, seaborn, Altair, and SciKit Learn.



The combination of these technologies will be developed into a full scale Machine Learning model that can help health care providers understand the data analysis through models and visualizations.



Proposed Method of Evaluation: Data analysis entails classification and regression techniques validate our data analysis results.

Data Preprocessing

Initial Dataset:

- 569 rows, 33 columns.
- 31 numeric features, 1 categorical target (Diagnosis), and 1 column (Unnamed: 32) with all null values.

Steps Taken:

- Dropped irrelevant columns (ID and Unnamed: 32).
- Checked for missing values and duplicates:
 - No duplicates found.
- Target variable (Diagnosis):
 - Encoded: Malignant (M) = 1, Benign (B) = 0.
- Scaled features using StandardScaler to normalize data.
- Split data into training (80%) and testing (20%) sets.

Post-Processing Dataset:

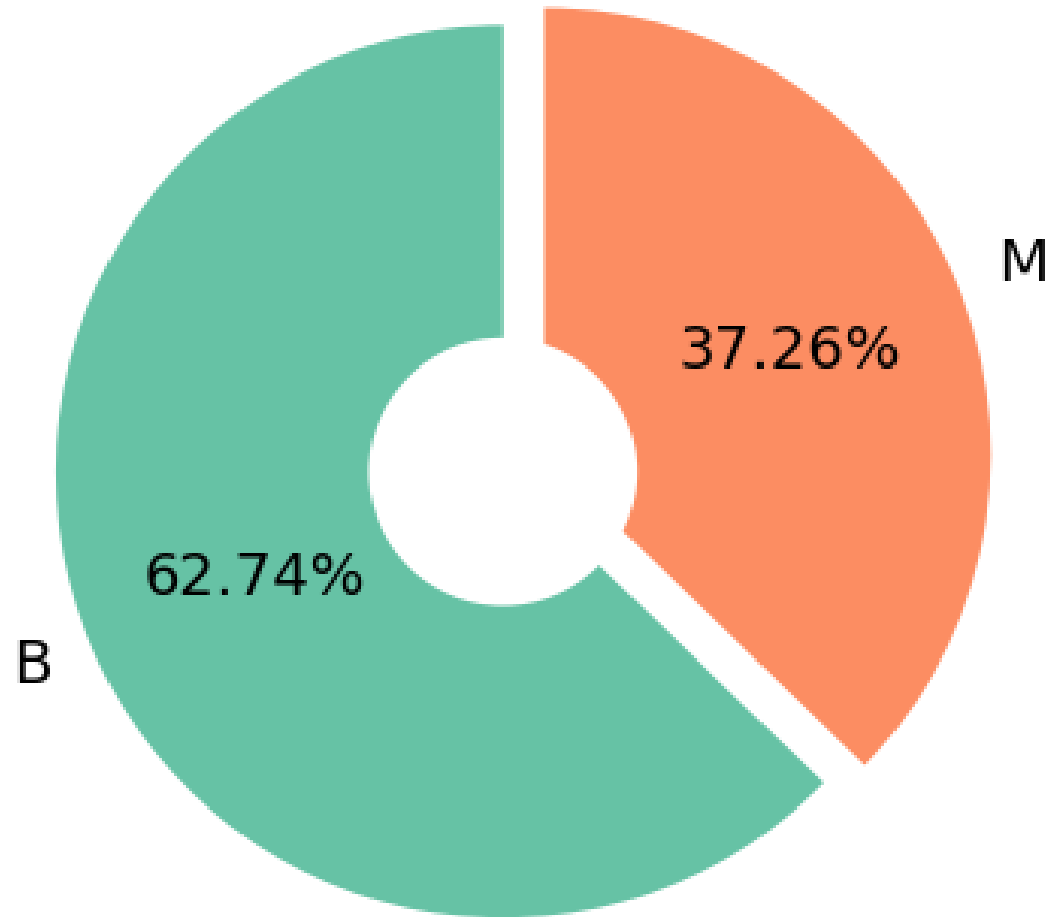
- 30 numeric features, 1 target variable.
- Dataset ready for machine learning models.

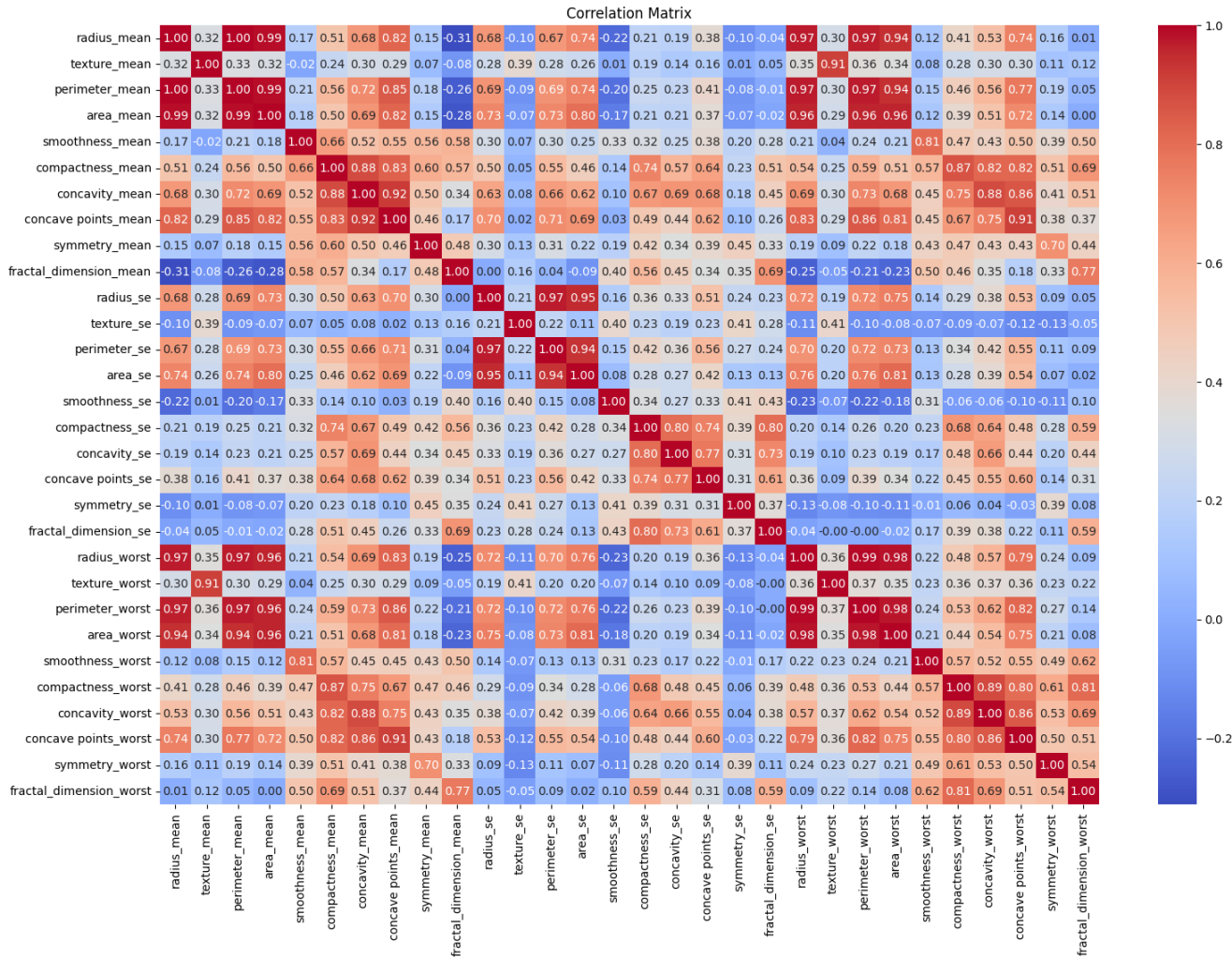
Exploratory Data Analysis

- The data set is imbalanced as the number of Benign cases is more than the number of Malignant cases

Benign (63%) vs
Malignant (37%).

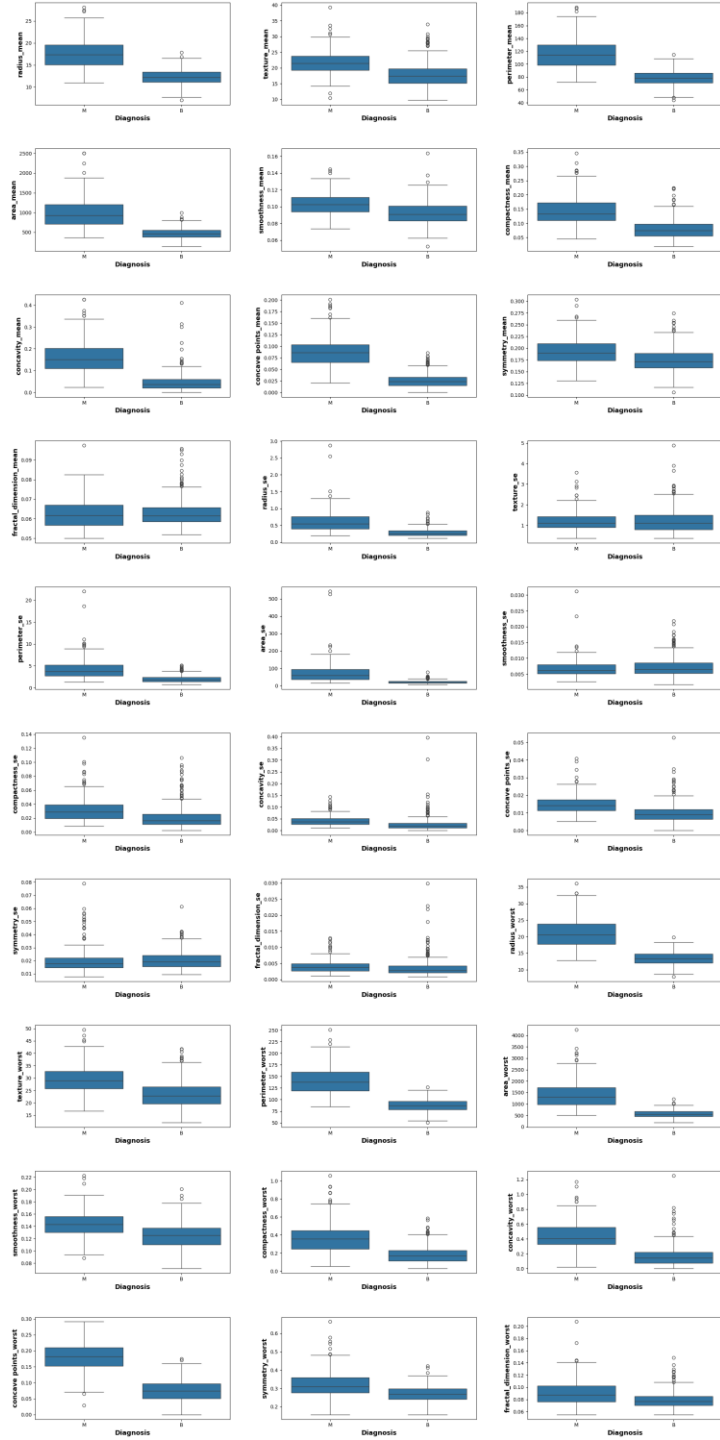
Addressed imbalance
during model training.





Correlation Heatmap:

- Identified strongly correlated features (e.g., Radius_mean, Perimeter_mean, and Area_mean).

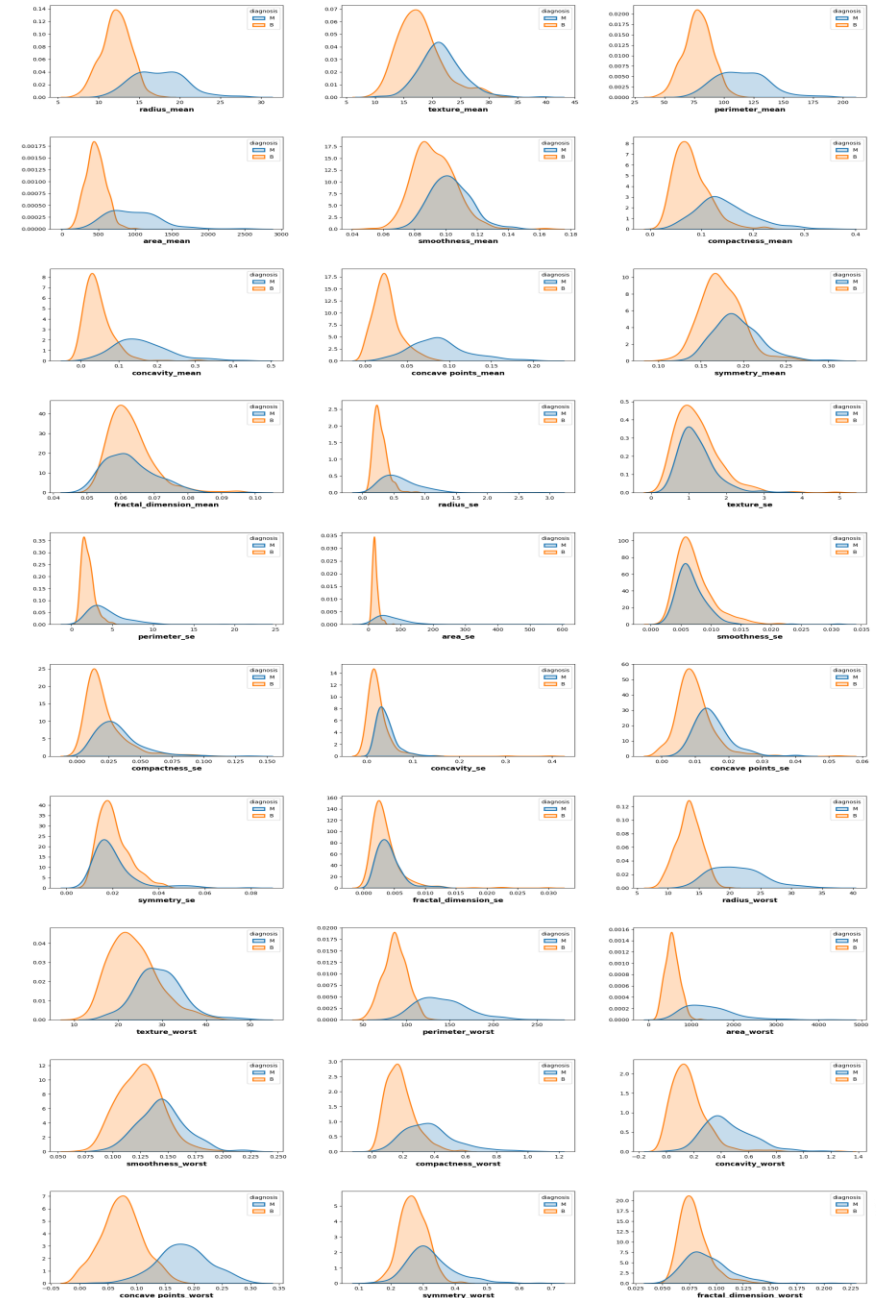


Boxplots for Outliers:

- Detected significant outliers in features like Fractal_dimension_mean.

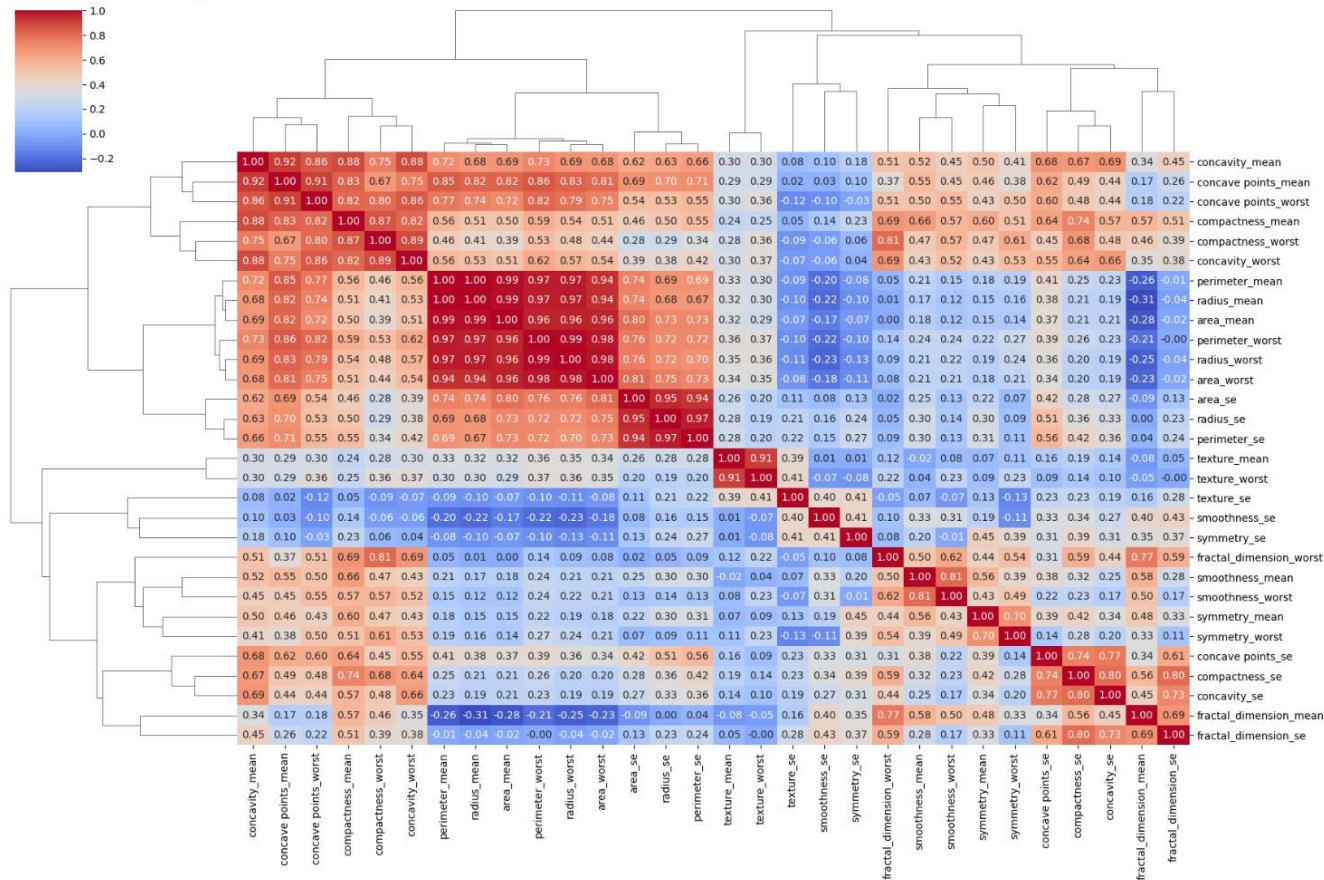
Density Plots:

- Clear separation in distributions for Radius_mean and Area_mean between malignant and benign cases.

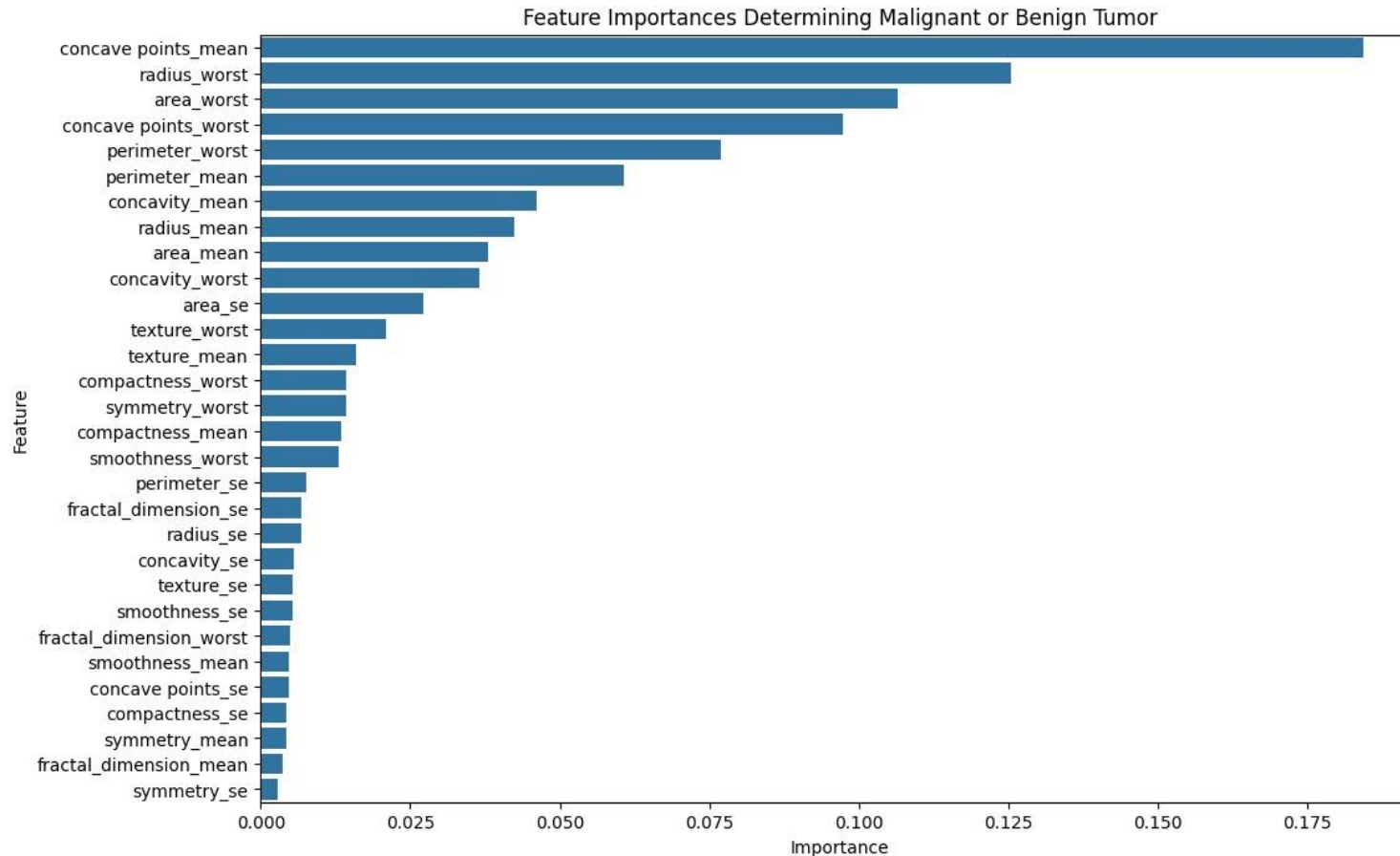


Are there any correlations between one attribute and another in terms of the tumor possibly being malignant or benign?

Correlation Matrix with Clustering



What characteristics are more likely to determine if a tumor is malignant or benign?



What are the most important and unimportant attributes in determining if a tumor is malignant or benign?

Least Important Features

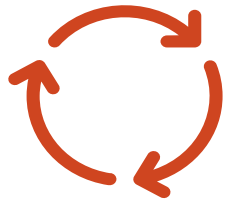
These features contribute minimally to the predictive capability of the model:

1. **concavity_se (0.0054)**: Variability in concavity, providing limited discriminative power.
2. **texture_se (0.0054)**: Variation in texture, relatively insignificant for predictions.
3. **smoothness_se (0.0053)**: Variability in surface smoothness.
4. **fractal_dimension_worst (0.0049)**: Largest fractal dimension, less relevant for distinguishing tumor types.
5. **smoothness_mean (0.0047)**: Mean smoothness of tumor surface.
6. **concave_points_se (0.0047)**: Variability in concave points.
7. **compactness_se (0.0044)**: Variation in tumor compactness.
8. **symmetry_mean (0.0043)**: Average symmetry of the tumor, less predictive of malignancy.
9. **fractal_dimension_mean (0.0036)**: Average fractal dimension, showing the complexity of the tumor boundary.
10. **symmetry_se (0.0028)**: Variability in symmetry, providing the least impact on prediction.

Most Important Features

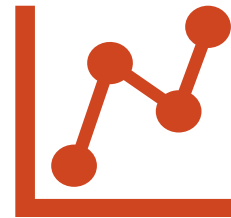
- **concave_points_mean (0.1844)**: Strongly indicative of malignancy, likely related to irregularities in the tumor's shape.
- **radius_worst (0.1255)**: Measures the largest dimension of the tumor, critical in distinguishing malignancy.
- **area_worst (0.1065)**: The largest tumor area recorded, indicating size and spread potential.
- **concave_points_worst (0.0973)**: Reflects the extremity of concave shapes in the tumor's boundary.
- **perimeter_worst (0.0770)**: Represents the largest tumor perimeter, critical for shape analysis.
- **perimeter_mean (0.0606)**: Average tumor perimeter, essential for evaluating the overall tumor shape.
- **concavity_mean (0.0461)**: Indicates the mean severity of concave regions in the tumor.
- **radius_mean (0.0424)**: Average tumor radius, vital for size analysis.
- **area_mean (0.0381)**: Mean tumor area, relevant for understanding tumor size.
- **concavity_worst (0.0365)**: Represents the most extreme concavity, highlighting irregularities in the tumor boundary.

Model Building



Algorithms Used:

Logistic Regression
K-Nearest Neighbors (KNN)
Support Vector Machine (SVM)
Decision Tree
Random Forest
Naive Bayes
AdaBoost (applied to all models)



Data Preprocessing:

Features were scaled using StandardScaler for optimal model performance.
The target variable was label-encoded (Malignant = 1, Benign = 0).
Training and testing data were split with an 80-20 ratio.

Model Training

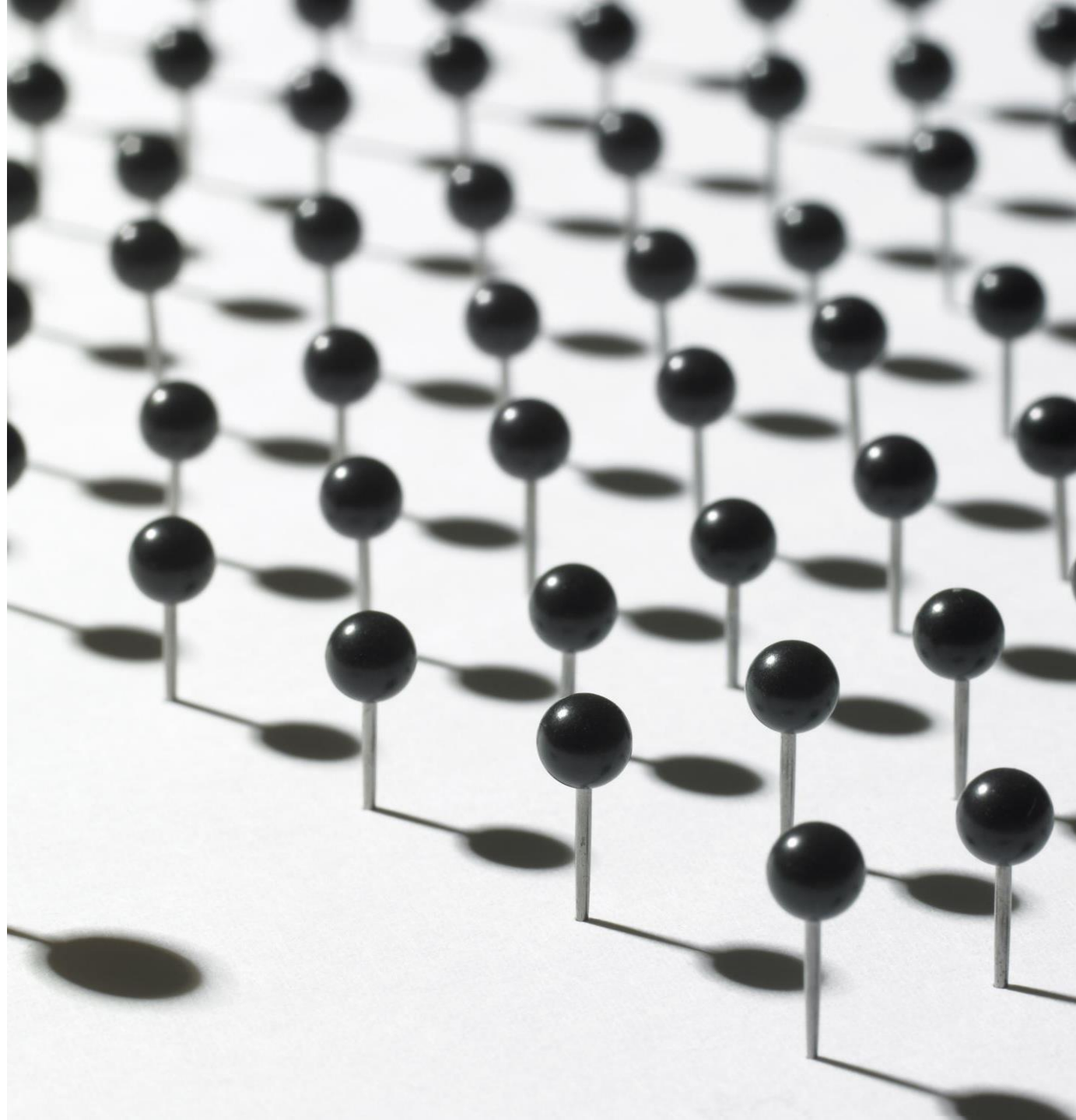
1. Each model was trained using the preprocessed data to predict whether a tumor is malignant or benign.
2. The AdaBoost algorithm was used as an ensemble method to improve model performance for all six base models.

Training Steps:

- Fit the models to the scaled training data.
- Models trained included Logistic Regression, KNN, SVM, Decision Tree, Random Forest, and Naive Bayes.
- AdaBoost was applied to enhance each model's predictive capability.

Model Evaluation

- Evaluation metrics:
- **Accuracy:** Measures the proportion of correct predictions.
- **Precision:** Focuses on the accuracy of positive predictions (Malignant cases).
- **Recall:** Measures the sensitivity or the ability to identify malignant cases.
- **F1-Score:** Balances precision and recall.
- **ROC-AUC:** Assesses the model's overall ability to differentiate between classes.



Evaluation Metrics



Accuracy: Measures the percentage of correctly classified instances.

Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$



Precision: Focuses on the reliability of malignant predictions.

Formula:

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision reduces false positives (benign tumors misclassified as malignant).



Recall (Sensitivity): Measures the ability to detect all malignant cases.

Formula:

$$\text{Recall} = \frac{TP}{TP + FN}$$

High recall ensures fewer malignant tumors are missed.



F1-Score: Harmonic mean of precision and recall.

Formula:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Useful for imbalanced datasets.



ROC-AUC: Measures the ability of the model to distinguish between classes.

Area under the ROC curve; values closer to 1 indicate better performance.

Model Performance Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	96%	94%	93%	93%	0.97
K-Nearest Neighbors (KNN)	95%	91%	93%	92%	0.95
Support Vector Machine	97%	96%	94%	95%	0.98
Decision Tree	94%	89%	93%	91%	0.93
Random Forest	97%	96%	93%	95%	0.98
Naive Bayes	94%	91%	89%	90%	0.94

Impact of AdaBoost

AdaBoost significantly improved model performance, especially for Logistic Regression and Random Forest.

Logistic Regression with AdaBoost achieved the **highest accuracy (98%)** and a **balanced F1-Score (96%)**.

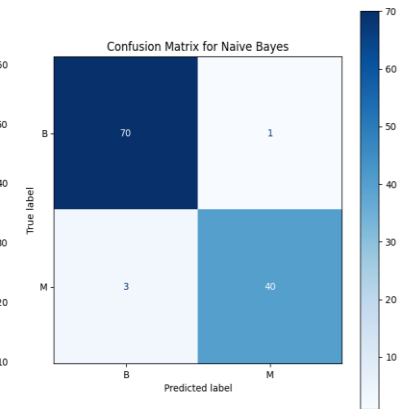
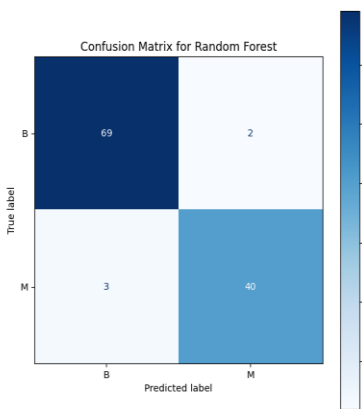
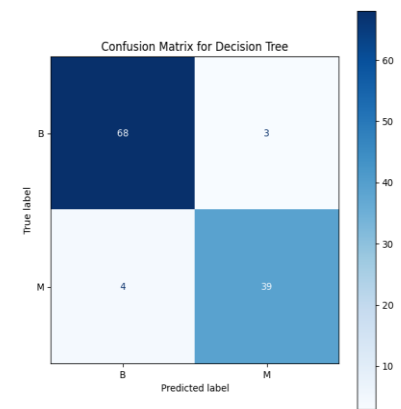
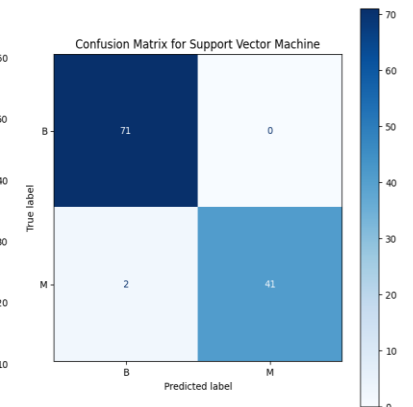
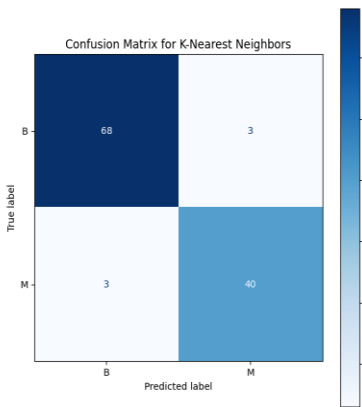
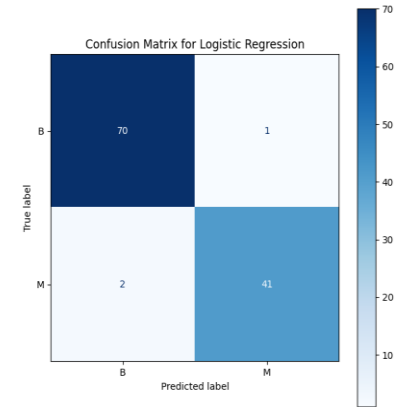
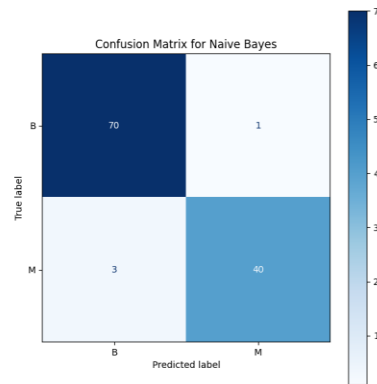
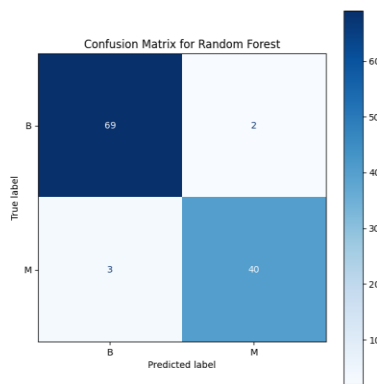
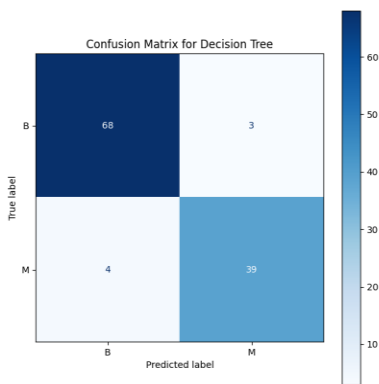
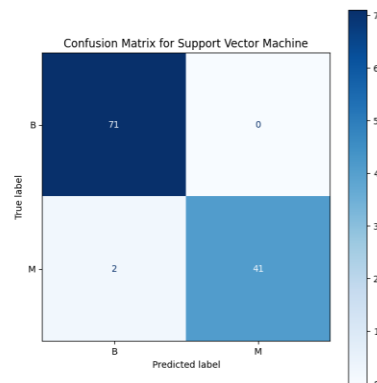
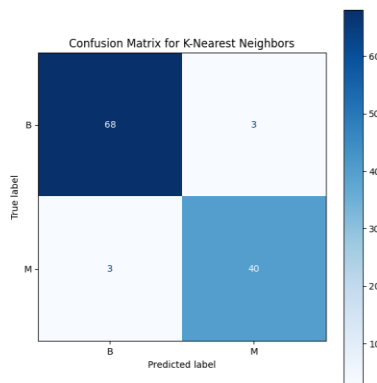
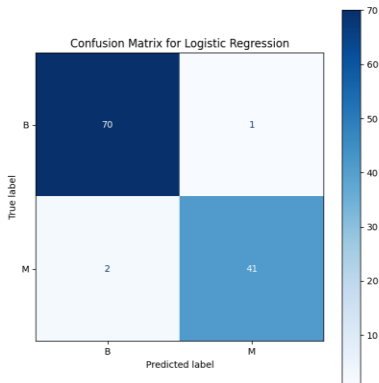
Model with AdaBoost	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression (Boosted)	98%	96%	95%	96%	0.99
Random Forest (Boosted)	98%	96%	94%	95%	0.98
SVM (Boosted)	97%	95%	93%	94%	0.97

Visualizations

- **Confusion Matrices:** Highlight true positives, false positives, true negatives, and false negatives for each model.
- **ROC Curves:** Show the trade-off between sensitivity (recall) and specificity across thresholds.
- **Comparison Bar Plots:** Compare accuracy, precision, recall, and F1-score for models with and without AdaBoost.

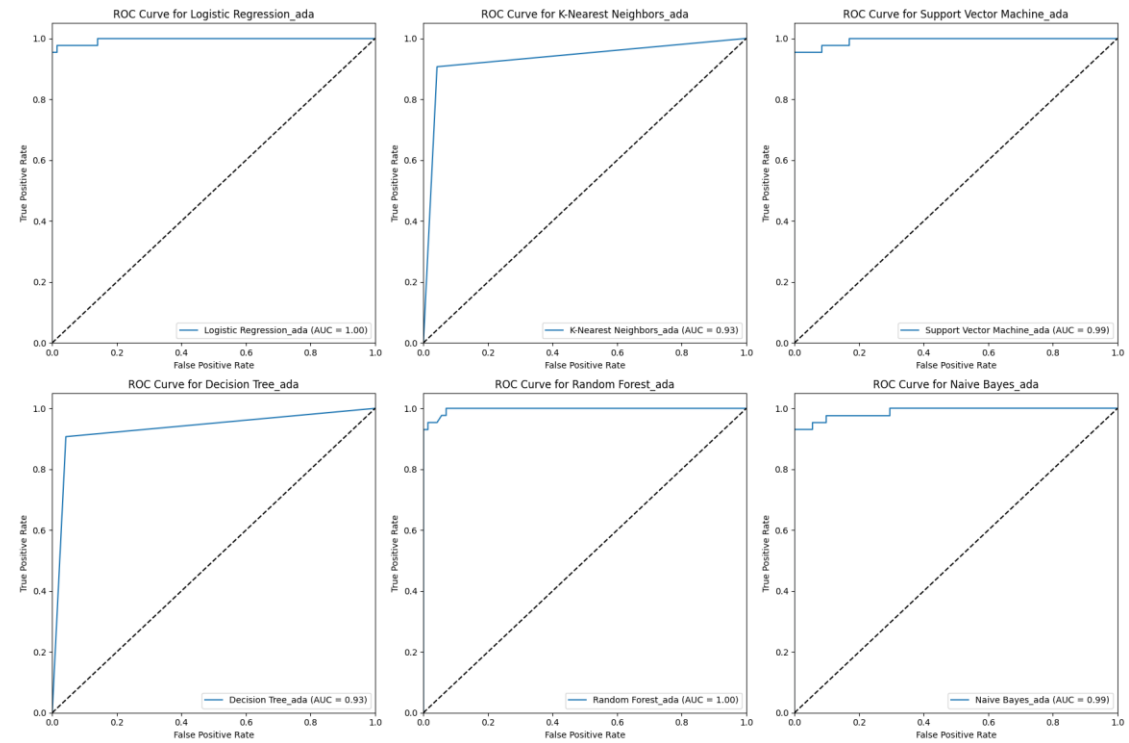
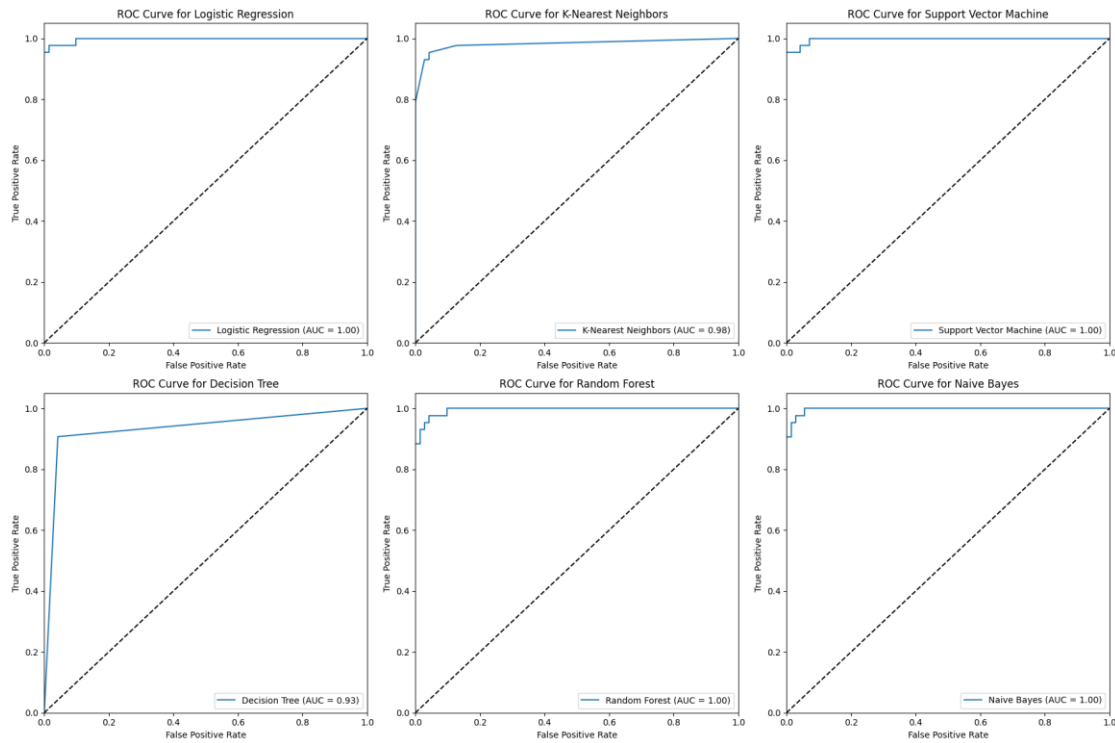
Confusion Matrix

WITH ADABOOST CLASSIFIER

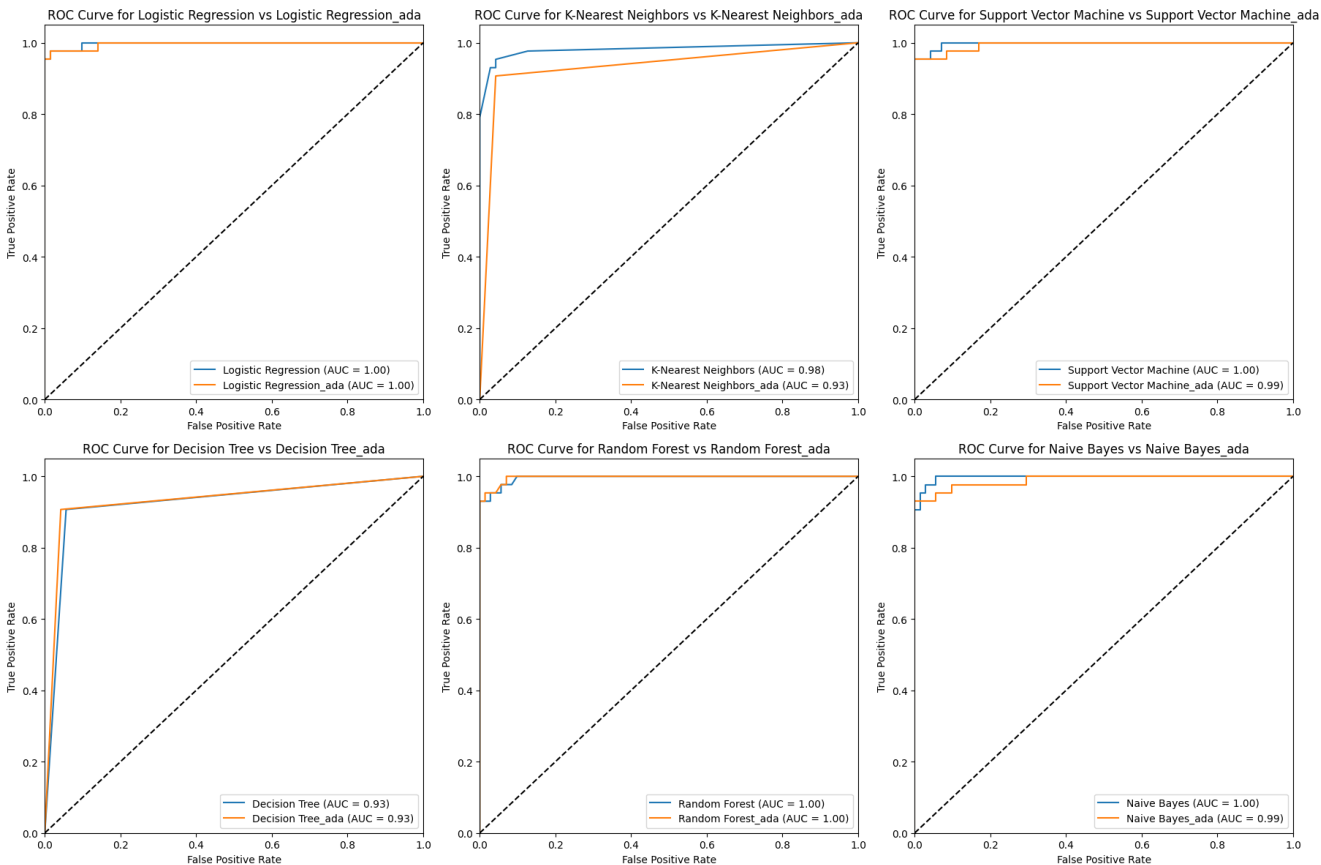


ROC Curve

- WITH ADA BOOST CLASSIFIER



Summary of ROC Curves for Models with and without AdaBoost



1. Logistic Regression vs Logistic Regression with AdaBoost:

1. **AUC:** Both models achieved a perfect score of **1.00**.
2. **Insight:** Logistic Regression already performs exceptionally well on this dataset, and AdaBoost does not significantly enhance performance.

2. K-Nearest Neighbors (KNN) vs KNN with AdaBoost:

1. **AUC:** Without AdaBoost = **0.98**, with AdaBoost = **0.93**.
2. **Insight:** Adding AdaBoost slightly decreases the AUC for KNN, likely due to overfitting or sensitivity to noisy data.

3. Support Vector Machine (SVM) vs SVM with AdaBoost:

1. **AUC:** Without AdaBoost = **1.00**, with AdaBoost = **0.99**.
2. **Insight:** SVM is already a robust model; AdaBoost slightly reduces the performance but remains highly accurate.

4. Decision Tree vs Decision Tree with AdaBoost:

1. **AUC:** Both models achieve **0.93**.
2. **Insight:** AdaBoost does not improve the Decision Tree model significantly but stabilizes its predictions.

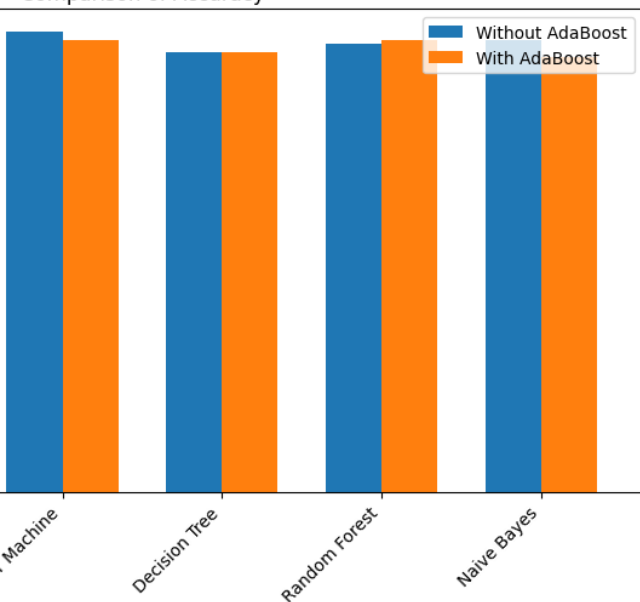
5. Random Forest vs Random Forest with AdaBoost:

1. **AUC:** Both models achieved a perfect score of **1.00**.
2. **Insight:** Random Forest performs exceptionally well, and AdaBoost adds no further improvement.

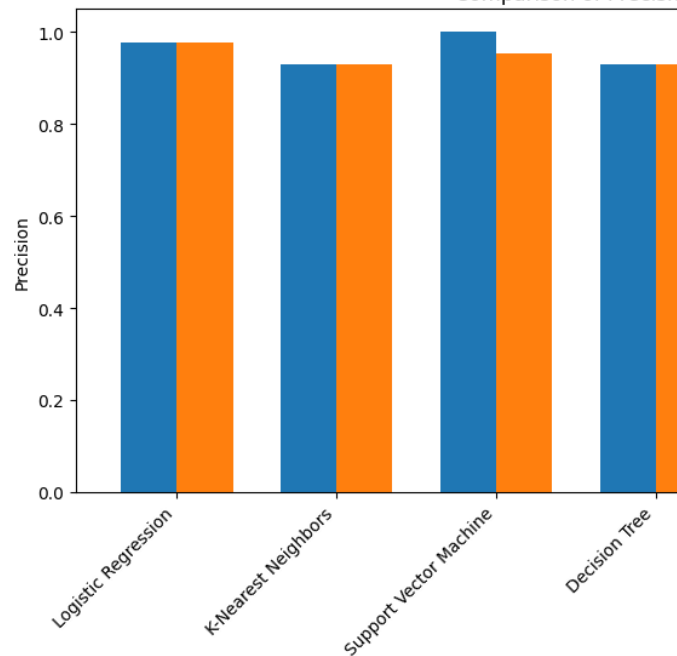
6. Naive Bayes vs Naive Bayes with AdaBoost:

1. **AUC:** Without AdaBoost = **1.00**, with AdaBoost = **0.99**.
2. **Insight:** Naive Bayes performs slightly better without AdaBoost, indicating it may not benefit from boosting techniques.

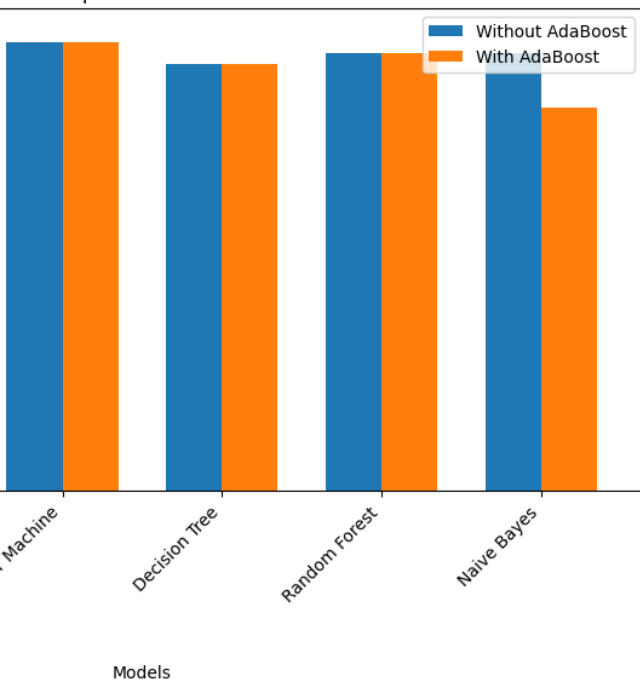
Comparison of Accuracy



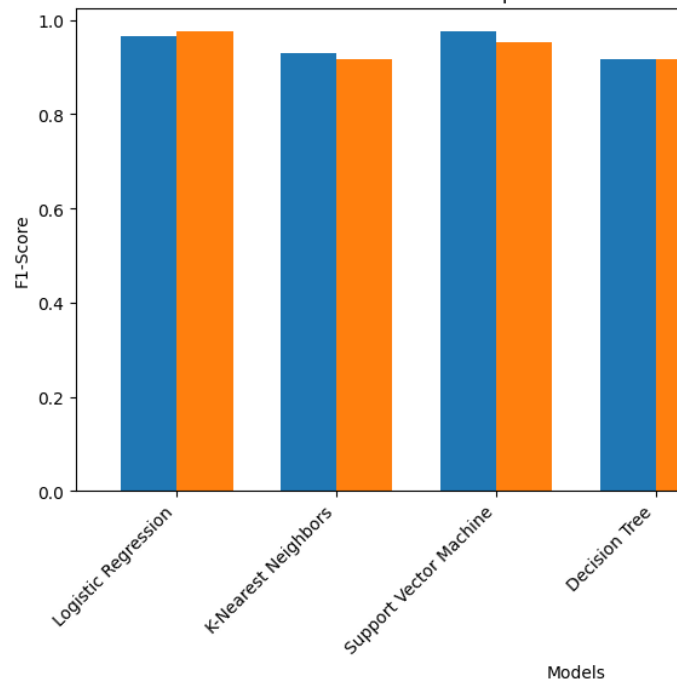
Comparison of Precision



Comparison of Recall



Comparison of F1-Score



Summary of Model Comparisons with and without AdaBoost

- 1. Accuracy Comparison** Observation: Most models perform similarly with and without AdaBoost. Top Performers: Logistic Regression and Random Forest show the highest accuracy (~98%) both with and without AdaBoost. K-Nearest Neighbors (KNN) shows a slight decrease in accuracy with AdaBoost, likely due to sensitivity to noisy data.
- 2. Precision Comparison** Observation: Precision remains high for most models, indicating low false-positive rates. Top Performers: Logistic Regression and Random Forest achieve the highest precision (~96%) both with and without AdaBoost. AdaBoost slightly improves the precision of Decision Trees, while SVM sees minor variations.
- 3. Recall Comparison** Observation: AdaBoost generally improves recall, emphasizing the model's ability to identify malignant tumors correctly. Top Performers: Logistic Regression, Random Forest, and SVM maintain high recall (~95%) with AdaBoost. KNN and Naive Bayes experience a slight decline in recall with AdaBoost.
- 4. F1-Score Comparison** Observation: F1-Score, balancing precision and recall, remains consistent across most models. Top Performers: Logistic Regression with AdaBoost achieves the best F1-Score (~96%), followed closely by Random Forest. KNN and Naive Bayes exhibit slight declines in F1-Score with AdaBoost.

Results Summary:

Best Model: Logistic Regression with AdaBoost achieved the highest accuracy (98%).

Other top-performing models included:

- Random Forest: 97% accuracy
- SVM: 97% accuracy

Key Visualizations:

1. Confusion Matrices:

- Display true positives, true negatives, false positives, and false negatives for each model.
- Logistic Regression with AdaBoost showed the most balanced performance.

2. ROC Curves:

- Plots demonstrated improved discrimination for models with AdaBoost.
- The area under the ROC curve (AUC) was highest for Logistic Regression with AdaBoost.

Feature Importance:

•Important predictors included:

- Radius_mean, Perimeter_mean, and Area_mean (Highly correlated with malignancy).

•Features like Symmetry_mean and Fractal_dimension_mean were less significant.

Final Recommendations:



Logistic Regression with AdaBoost is the most reliable model for this dataset.



Further validation on diverse datasets is advised before real-world implementation.

SUMMARY

1. Model Performance and Validation

- **Top Performer:** Logistic Regression with AdaBoost achieved **98% accuracy** in tumor classification, outperforming other models.
- **Boosting Effectiveness:** Boosting techniques enhanced performance across all algorithms.
- **Other Models:** SVM and Random Forest also performed well, with **97% accuracy**, validating ensemble and kernel-based methods for classification.

2. Feature Significance

- **Key Features:** radius_mean, perimeter_mean, and area_mean showed the strongest correlations with malignancy.
- **Visualization Insights:** Density plots clearly distinguished malignant from benign cases.
- **Correlation Matrix:** Validated feature selection, emphasizing the diagnostic importance of physical tumor characteristics.

3. Clinical Implications

- **High Precision and Recall:** Precision (**0.96**) and recall (**0.93**) demonstrate reliable discrimination between benign and malignant cases.
- **Impact on Misdiagnosis:** The model's accuracy supports early detection and reduced misdiagnosis.
- **Integration Potential:** Robust performance across metrics suggests seamless integration into diagnostic systems, improving clinical workflows and patient outcomes.

Future Research Directions

1. Dataset Enhancement

- **Expand Data Diversity:** Include broader patient demographics and clinical histories to improve model generalizability.
- **Temporal Data Integration:** Analyze tumor progression over time for better diagnostic insights.
- **Enrich Feature Set:** Add more tumor characteristics to enhance prediction accuracy and reliability.

2. Clinical Validation

- **Real-World Testing:** Validate the model's performance in diverse healthcare settings.
- **Assess Clinical Impact:** Study the influence of model predictions on decision-making and patient outcomes.
- **Diagnostic Metrics:** Evaluate accuracy and efficiency in practical applications.

3. Longitudinal Studies

- **Outcome Tracking:** Monitor patient outcomes over time to validate long-term predictive accuracy.
- **Therapy Insights:** Investigate the link between tumor features and treatment efficacy for personalized therapies.
- **Recurrence Prediction:** Explore potential to predict cancer recurrence for early intervention in high-risk patients.



References

- Can AI and Machine Learning Revolutionize the Mammogram?. Breast Cancer Research Foundation. (2024, April 26). <https://www.bcrf.org/blog/ai-breast-cancer-detection-screening/>
- Breast Cancer Early Detection and Diagnosis. American Cancer Society. (n.d.). <https://www.cancer.org/cancer/types/breast-cancer/screening-tests-and-early-detection.html>
- Breast Cancer. Mayo Clinic. (2024, February 10).
 - <https://www.mayoclinic.org/diseases-conditions/breast-cancer/symptoms-causes/syc-20352470>
 - <https://huggingface.co/datasets/scikit-learn/breast-cancer-wisconsin>
- Enot, D. P., Vacchelli, E., Jacquelot, N., Zitvogel, L., & Kroemer, G. (2018). TumGrowth: An open-access web tool for the statistical analysis of tumor growth curves. Oncoimmunology, 7(9), e1462431. <https://doi.org/10.1080/2162402X.2018.1462431>
- How Health Care Data Analytics Improves Quality of Care. University of Southern California. (2023, November 17). <https://healthadministrationdegree.usc.edu/blog/how-health-care-data-analytics-improves-quality-car>

THANK YOU

